

Request for Proposal — GhostLink GDL v1 VM

Project

GhostLink — GDL v1 VM Implementation (Fixed-Bid)

Date

October 03, 2025

Background

GhostLink is a difference-first substrate language (GDL v1) and Proof-of-Record (PoR) suite. We need a compact virtual machine (VM) that executes the 16-opcode IR and passes the PoR gates.

Objectives

- Implement a tiny high-performance VM for GDL v1 (CPU: x86-64 or ARM64).
- Execute the provided example program end-to-end.
- Produce JSON telemetry for PoR verification.
- Achieve determinism tier D1 (seeded per-cell RNG).

Scope of Work

- 1) Implement 16 opcodes (see `gdl_ir_opcodes.json`).
- 2) SoA memory layout; active-set (difference-only) execution; CSR/stencil neighbors.
- 3) CLI runner: ``gdllrun program.gdl --steps K``.
- 4) Telemetry: activity ratio, Σ mass over time, avalanche sizes, MDL proxy, anomaly index A, outcome C.
- 5) Export legacy log for origin-finding.
- 6) Docs: build + opcode mapping + determinism rules.

Deliverables

- Static or shared library (``libgdlvm``) and CLI runner.
- Telemetry JSON + example outputs.
- Build/documentation + test instructions.
- Source code and rights assignment.

Acceptance Criteria (PoR)

- Difference-only cost: avg active ratio ≤ 0.20 and $\leq 0.30\times$ cost vs full sweep.
- Predictive lift: $\geq 5\%$ vs neighbors-only baseline.
- SOC: avalanche tail exponent $1 < \tau < 3$ (KS $p \geq 0.05$), branching ≈ 1 .
- Reproducibility: stdev/mean $\leq 10\%$ across 5 seeds.
- Topology invariance: metrics within 10% across planar vs spherical.
- Ablation sanity: removing scars/self/order degrades $\geq 15\%$.

Timeline

Week 0: Kickoff
Week 1: Architecture & scaffolding
Week 2: Core opcodes + runner
Week 3: PoR metrics + tuning
Week 4: PoR pass + docs/handoff

Budget Guidance

Target fixed-bid ranges (USD):

- Senior individual: 12k-22k (4-5 weeks, 1 developer)
- Small shop: 18k-35k (parallelize PoR/tests)

Propose milestones with payment schedule.

Submission Requirements

- Short plan (≤ 2 pages) describing approach, perf plan, and determinism strategy.
- Relevant repos or code samples (VMs, interpreters, SIMD).
- Fixed bid, schedule, and milestone breakdown.
- Hourly rate for out-of-scope changes.
- Availability window.

Evaluation Criteria

Weighting: Technical fit (35), Performance history (20), Testing discipline (15), Communication (10), Cost (10), Availability (10).

References

Artifacts included: GhostLink_Final_Project.pdf, GDL_v1_Substrate_Language.pdf, GDL_v1_VM_Implementors_Checklist.pdf, gdl_ir_opcodes.json, gdl_example.gdl, gdl_example.ir.json.